

```

1 #include <WiFi.h>
2 #include <AsyncTCP.h>
3 #include <ESPAsyncWebServer.h>
4
5 #define RADAR_PIN 4 // HB100 IF output connected to GPIO4
6
7 // Radar variables
8 volatile unsigned long pulseCount = 0;
9 unsigned long lastMillis = 0;
10 unsigned long pulsePerSecond = 0;
11 float speed_mps = 0;
12 float speed_kmph = 0;
13 float threshold_speed = 10.0; // default threshold in km/h
14
15 // Web Server
16 AsyncWebServer server(80);
17 AsyncWebSocker ws("/ws");
18
19 // AP Credentials
20 const char* ssid = "IndusRadar";
21 const char* password = "12345678";
22
23 const char htmlPage[] PROGMEM = R"rawliteral(
24 <!DOCTYPE html>
25 <html>
26 <head>
27 <meta charset="UTF-8">
28 <title>IndusSpeed Radar</title>
29 <style>
30 body {
31 background: #ffffff;
32 color: #222222;
33 font-family: "Segoe UI", sans-serif;
34 margin: 0;
35 padding: 0;
36 text-align: center;

```

Fig. 3: Code snippet configuring Wi-Fi

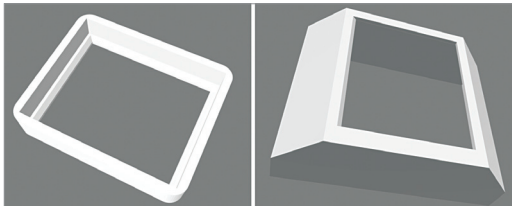


Fig. 4: A proposed radar enclosure design

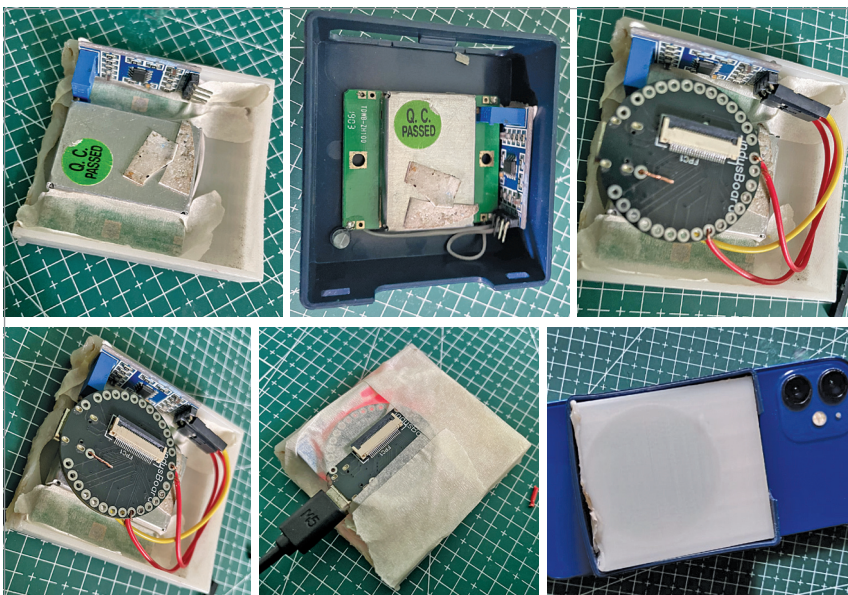


Fig. 5: Different fixing stages of the radar in the enclosure

BILL OF MATERIALS

Components	Quantity	Description
IndusBoard Coin (ESP32-S2)	1	Microcontroller board with built-in Wi-Fi
HB100 Doppler radar sensor	1	Measures speed based on the Doppler effect
LM358 amplifier module	1	Amplifies HB100 signal for better detection
USB Type-C cable	1	For power and programming
3.3V battery / 3.3V-5V DC	1	For power supply
An enclosure for the device	1	For its protection against adverse weather

paired with mobile devices.

The Bill of Materials table above details the required components, while Fig. 1 shows the author's assembled prototype being tested on a tripod. The system can even be mounted on a roadside pole for smart city applications and continuous monitoring.

Understanding the Doppler effect in radar

The Doppler effect is a fundamental principle behind modern radar speed detection systems. It refers to the change in the frequency and wavelength of waves, such as sound or radio waves, caused by the relative

motion between the wave source and the observer.

Two main scenarios illustrate this phenomenon with sound:

1. Moving source, stationary observer. If the sound source moves toward the observer, the sound waves become compressed, resulting in a higher observed frequency (a higher-pitched sound). Conversely, the sound waves stretch out when the source moves away, causing a lower observed frequency (a lower-pitched sound).

2. Stationary source, moving observer. If the observer moves toward the stationary source, the perceived frequency increases. If the observer moves away, the perceived frequency decreases, producing a similar Doppler effect.

Radar-based speed detection systems operate on the same core principle, but instead of sound waves they use radio frequency (RF) signals, typically in the microwave range.

How Doppler radar works

Doppler radar operates by transmitting RF signals in a specific direction using antennas or sensors. These are typically mounted on a vehicle's front grille or bumper, or externally on poles or portable systems. The emitted signals travel outward, scanning the surroundings for potential obstacles, vehicles, or pedestrians.

When these signals encounter an object, a portion of the signal is reflected back to the radar receiver. If the object is stationary or moving at the same speed as the radar,